

Education

University College London

September 2023 – September 2024

MSc Machine Learning (*Distinction*)

Probabilistic and Unsupervised Learning¹ | Approximate Inference¹ | Bayesian Deep Learning
Applied ML | Supervised Learning | Reinforcement Learning² | Applied Deep Learning

MSc Thesis (*Distinction*) - supervised by Prof. Mirco Musolesi.

Hierarchical Bayesian Program Synthesis for Neural Algorithmic Reasoning

Imperial College London

October 2020 – June 2023

BSc Physics with Theoretical Physics

BSc Thesis (*Distinction*) - supervised by Dr David Clements. *Computational Evolution*

Experience

Machine Learning Consultant — Stealth Startup (Consulting), London

2025–

Feasibility study on LLM integration to company's core product for survey schema detection.
Implementation, documentation of methodology and experiments, strategic recommendations.

Researcher — University College London

2024–

Research on Neural Program Synthesis and Algorithmic Reasoning in the Machine
Intelligence Lab, supervised by Prof. Mirco Musolesi.

Research Intern — Imperial College London

2023

Causal discovery using cross-validation as compared with Bayesian Model Selection (using
GPLVM), supervised by Mark van der Wilk and funded by Imperial UROP. HPC experience.

Projects

Transformer Program Synthesis — University College London

2025

Developed novel programming-by-example framework: Transformer-based model recovers
algorithms from data, training neural functions on-the-fly.

Masked Image Modelling — University College London

2024

Self-supervised learning framework: VisionTransformer pre-trained to reconstruct partially
masked images and fine-tuned on Oxford-IIIT Pets dataset for segmentation. (*Distinction*)

Machine learning for particle physics — Imperial College London

2022

Lead ML team analysing decay product distribution from a Large Hadron Collider experiment
(LHCb) for Standard Model comparison. TensorFlow, XGBoost. (*Distinction, highest score*)

Solar Sail (Project Svarog) — Imperial College Space Society

2022

Orbital trajectory simulation, atmospheric effects, relativistic corrections. Launching in 2028.

Honours & Achievements

2nd Year Quantum Physics Article Prize – Published on Imperial Website (2022)

Young Researchers of Natural Sciences, Conference Presentation Distinction (2016)

Skills & Languages

Python (PyTorch, TensorFlow, GPflow, JAX, OpenAI API), Julia, parallelisation on GPU cluster

Hungarian (Native), English (Proficient - Cambridge Advanced 206), French (Intermediate)

¹Module part of PhD programme at the Gatsby Computational Neuroscience Unit

²Module taught by Google DeepMind

For references please contact:

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